



SoC 2026 - Face Aging Model

1. Deep Learning Basics

1.1. Deep Learning Vs Statistical ML

So, basics of any ML Model is to predict some output after going through identification of pattern and trends in some given huge amount of dataset. It can be a text, image, video or anything else also possible. Depending on Algorithm used, kind of Data trained and mainly the basic method adopted by the model to undergo training and do the output prediction, there can be 2 main kinds of ML -

- Deep Learning
- Statistical ML

Deep Learning	Statistical ML
Learns features automatically	Requires manual feature engineering
Needs Large Datasets	Works with smaller datasets
Uses Neural Networks	Uses traditional ML algorithms
Best for images, audio, text	Best for structured/tabular data
Higher computational cost	Lower computational cost

1.2 What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses **Artificial Neural Networks (ANNs)** with multiple layers to learn patterns directly from data.

Instead of manually defining features, the model learns useful features on its own during training.

Key Advantages

- Automatic feature extraction

- Handles complex data efficiently
- High performance on image tasks
- Improves with more data

1.3 Artificial Neural Networks (ANN)

ANNs are the basic building blocks of Deep Learning.

A neuron takes inputs, performs computations and passes the output forward.

$$z = \sum w_i x_i + b$$
$$a = f(z)$$

where w are weights, b is bias and f is an activation function.

1.4 Training and Inference

Training: Model learns patterns from labelled data by minimizing prediction error.

Inference: Trained model is used to generate outputs for new unseen images.

For Face Aging, training helps the model learn age transformations, while inference generates the aged version of a new face image.